

WE FOCUS ON TWO ONE-SIDED BEHAVIORAL TRANSITIONS

Taxonomy turns aggregate model differences into semantically targeted cohorts

DEEPSEEK · 215 IMPROVEMENTS

base fail → fine-tuned pass

119/215: generated test/import contamination
48: wrong output or logic
16: missing name/import
32: other primary categories

Mechanistic focus: 80 contamination cases

CODELLAMA · 291 REGRESSIONS

base pass → merged fail

120/291: API/type mismatch
50: wrong logic/other runtime
43: edge case/exception
78: generation, import, syntax, commentary

Mechanistic focus: 50 wrong-logic/runtime cases
Improvements: only 4 positives · EXCLUDED

HOW WE CLASSIFIED

Rule/AI-assisted taxonomy + human spot checks

IMPORTANT QUALIFICATION

Primary = most salient observed error—not a unique root cause. CodeLlama failures are heterogeneous and oft

DEEPSEEK /5 · CONTAMINATION

BASE · FAIL

Correct-looking body continues into:

```
`#tests/test_task_func.py`  
`import pytest`
```

FINE-TUNED · PASS

Completes the function and stops after a short example.

First textual divergence $\approx 76\%$ of shorter code;
target boundary is late/post-solution.

CODELLAMA /490 · WRONG LOGIC/API CONTRACT

BASE · PASS

parses XML $\rightarrow$ writes JSON file $\rightarrow$ returns result

MERGED · FAIL

```
`return xmltodict.parse(s)`  
omits the required file write side effect
```

First divergence $\approx 81\%$ of base code;
target region is the late function body.

ERROR-FOCUSED TOP-10 LISTS DEFINE THE NEW $\alpha=3$ CANDIDATE POOL

Three summaries per cell: max · mean · active fraction; ranking by absolute E/V

DEEPSEEK · 80 CONTAMINATION IMPROVEMENTS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
13879	14175	2913	2468
6895	9666	7728	14295
1078	1206	15821	8661
596	8686	12308	7728
8185	7169	16264	2913
14820	11612	2957	2957
1939	7804	3602	9700
15235	1939	2621	6305
15733	4994	6877	10736
16205	1591	14006	3602

40 nominations → 35 unique features

5 feature(s) appear in more than one list

CODELLAMA · 50 WRONG-LOGIC REGRESSIONS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
7692	15559	8875	4173
10818	3097	7588	15559
5642	565	12895	12623
447	4173	447	15968
11596	15890	4634	8142
1017	414	5862	9419
4634	4740	3366	13231
7588	13768	7486	414
3442	4309	1017	13151
8384	14969	5642	13172

40 nominations → 32 unique features

8 feature(s) appear in more than one list

NEW $\alpha=3$ SWEEP: exactly 35 DeepSeek + 32 CodeLlama unique features · no post-hoc additions

Hard gate: baseline must reproduce the original raw completion byte for byte before any steered arm runs.